

Sriram Sekhar Bharadwaj

📍 University of California, Los Angeles ✉ sriramsb@ucla.edu 🌐 [Website](#) 🎓 [Google Scholar](#)

Education

University of California, Los Angeles

Sept 2021 – Present

PhD in Theoretical Physics

- **PhD advisors:** Eric D'Hoker and Leo Zhou, **Mentors:** Di Luo and Zhong-Bo Kang
- **Awarded**
 1. “UCLA Physics & Astronomy Graduate Research Award” (2026-27), amount: \$30k
 2. “UCLA Robert Finkelstein Fellowship in Theoretical Physics” (2022), amount: \$6k
 3. “Mani L. Bhaumik Graduate Fellowship in Theoretical and Quantum Physics” (2023-26), amount: \$6k/summer
- **Semi-finalist** (top 20 of 133) in the **XPrize Quantum Applications** competition (2025): “CV-QAOA: Efficient Low-Depth Quantum Optimization for Continuous Variables” – with Di Luo and Leo Zhou
- **Scientific computing skills:** tensor networks/DMRG using Julia and iTensor, exact diagonalization, variational quantum algorithms, numerical optimization, neural quantum states, Python/SciPy/PyTorch etc.

University of Oxford

Oct 2020 – Sept 2021

MSc in Mathematical and Theoretical Physics

- Graduated with distinction and completed a double-unit master's dissertation on string amplitudes

University of California, Los Angeles

Sept 2017 – Sept 2020

BS in Mathematics and Applied Science

- Graduated *summa cum laude* and was on Dean's honors list every quarter

Publications (in reverse chronological order)

1. **(In Preparation) CV-QAOA: Efficient Low-Depth Quantum Optimization of Continuous Variables** Sep 2026
(anticipated)
with Di Luo and Leo Zhou, [View Draft](#)
2. **Confinement Versus Screening in the Schwinger Model on AdS_2 from Bosonization and Tensor Networks** Jul 2026
with Zhong-Bo Kang, Jack Isen, [arXiv:2607.19468](#)
3. **Information-Theoretic Scaling Laws of Neural Quantum States** Mar 2026
with Y. Lu, D. Rathore, and Di Luo, [arXiv:2603.23468](#)
4. **Hamiltonian Lattice QED₃ with One and Two Flavors of Wilson Fermions: Topological Structure and Response** Mar 2026
with Di Luo, Lena Funcke, Karl Jansen, et al., accepted [Physical Review D](#); [arXiv:2603.05616](#)
5. **A Path to Quantum Simulations of Topological Phases: (2+1)D Quantum Electrodynamics with Wilson Fermions** Apr 2025
with Di Luo, Lena Funcke, Karl Jansen, et al., accepted [Physical Review Letters](#); [arXiv:2504.21828](#)
6. **Non-Invertible Defects on the Worldsheet** Sep 2024
with P. Niro and K. Roumpedakis, published in [JHEP 03 \(2025\) 164](#)
7. **Approaching Argyres-Douglas Theories** Oct 2023
with Eric D'Hoker, published in [JHEP 06 \(2024\) 082](#)

Selected ongoing research (in order of planned submission)

Exponentially Accurate Adiabatic State Preparation for Lattice Gauge Theories and Bosonic Systems

with Di Luo

- Proves adiabatic theorem for truncated unbounded Hamiltonians with errors exponentially small in runtime uniformly in the cutoff; applied to $(2+1)\text{D}$ $\text{U}(1)$ and $\text{SU}(2)$ lattice gauge theories and the Hubbard-Holstein model

- Improved scaling with cutoff over previously considered algorithms for $(2 + 1)$ D lattice gauge theories, super-polynomial improvement over state-of-the-art classical methods for the Hubbard-Holstein model

Simulating Topological Phases of QED₃ on a Quantum Computer Using VQE

with Di Luo et al.

- VQE to prepare topological ground states of QED₃ with rigorous gate-complexity estimates
- Ongoing experimental validation of the predicted topological phase diagram of [2504.21828](#)  and [2603.05616](#)  on superconducting quantum hardware

List of presentations (in reverse chronological order)

- **(Upcoming) Rising Researchers Seminar:** Institute for Nuclear Theory, UW Seattle (Oct 2026)
- **(Upcoming) HEP-QIS Seminar:** Lawrence Berkeley National Laboratory, Host: Christian Bauer (Oct 2026)
- **Seminar:** “Quantum Simulations of Topological Phases: $(2+1)$ D Lattice QED with Wilson Fermions”, University of Illinois Urbana-Champaign, Host: Patrick Draper (Aug 2026)
- **High-Energy Theory Seminar:** “Quantum Simulations of Topological Phases”, Argonne National Laboratory, Host: Kyle Lee (Aug 2026)
- **Nuclear Theory/QIS Seminar:** Center for Frontiers in Nuclear Theory, Stony Brook University, Host: Felix Ringer (Aug 2026)
- **Seminar:** “Quantum Simulations of Topological Phases: $(2+1)$ D Lattice QED with Wilson Fermions”, Boston College, Host: Xiao Chen (Aug 2026)
- **Contributed talk:** “Quantum Simulations of Topological Phases: $(2+1)$ D Lattice QED with Wilson Fermions” (arXiv: [2603.05616](#) ) , IAIFI Workshop, MIT (Aug 2026).
- **Contributed talk:** “Quantum Simulations of Topological Phases: $(2+1)$ D Lattice QED with Wilson Fermions” (arXiv: [2603.05616](#) ) , The 43rd International Symposium on Lattice Field Theory (Lattice 2026), University of Maryland, College Park (Jul 2026).
- **Poster:** “Quantum Simulations of Topological Phases in $(2+1)$ D Lattice Gauge Theory” (arXiv: [2504.21828](#) ) , “QuantHEP 2025”, Lawrence Berkeley National Laboratory (Oct 2025).
- **Lightning talk:** “Quantum Simulations of Topological Phases in $(2+1)$ D Lattice Gauge Theory” (arXiv: [2504.21828](#) ) , NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Summer School, Harvard (Aug 2025).
- **Contributed talk:** “Probing the frontiers of nuclear physics with AI at the EIC (II)”, Stony Brook (Mar 2025).
- **Gong-show talk:** “Non-invertible symmetries on the worldsheet” based on [JHEP 03 \(2025\) 164](#) , “2024 IHES Summer School - Symmetries and Anomalies: a Modern Take”, IHES, France (June 2024).
- Pedagogical presentations in the High-Energy Theory Journal Club at UCLA.
 1. Confinement vs. screening in QED₂ on AdS₂ (2026)
 2. Anomalies and instantons in gauge theory (2024)
 3. Liouville field theory and 2D quantum gravity (2023)
 4. Quantum information theory and entanglement entropy in gravity (2023)
 5. Aspects of confining strings in gauge theory (2022)

Teaching and Professional Experience

- Refereed a paper submitted to the Quantum Information Processing (QIP) conference, 2025
- Was a teaching assistant for 10+ undergraduate courses, both elementary and advanced
- Appointed as lead teaching assistant to manage and update a course with over 900 students

List of conferences and schools (in reverse chronological order)

- The 43rd International Symposium on Lattice Field Theory (Lattice 2026), UMD, College Park, 2026
- Institute for AI and Fundamental Interactions Workshop 2026, MIT
- Institute for AI and Fundamental Interactions Summer School 2026, University of Massachusetts, Boston
- Quantum Winter School 2026: Quantum Simulation at Institute for Pure and Applied Mathematics, UCLA
- Institute for AI and Fundamental Interactions Summer School and Workshop 2025, Harvard University
- 2024 Summer School on Symmetries and Anomalies, IHES, France
- 50 years of QCD, UCLA

References

- Leo Zhou, Assistant Professor of Electrical and Computer Engineering, UCLA, ✉ leoxzhou@ucla.edu
- Di Luo, Associate Professor of Physics, Tsinghua University (formerly UCLA ECE), ✉ diluo@tsinghua.edu.cn
- Zhong-Bo Kang, Professor of Physics, UCLA, ✉ zkang@physics.ucla.edu
- Eric D'Hoker, Distinguished Research Professor in Theoretical Physics, UCLA, ✉ dhoker@physics.ucla.edu